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7 September 1956
SUPERSEDING
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MILITARY SPECIFICATION
SURFACER (SANDER), COMBINATION, BELT AND DISC, FLOOR MODEL

This specification has been approved by the Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 This specification covers one type of surfacer which is a bench type combination belt and disc sanding machine complete and assembled to a stand for floor mounting.

2. APPLICABLE DOCUMENTS

2.1 General.- The following specifications, standards, drawings and publications of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

TT-E-485	Enamel; Semi gloss, Rust-Inhibiting
TT-C-595	Colors; (for) Ready Mixed Paints

Military

JAN-P-72	Primer, Synthetic, Lacquer-Resisting
MIL-B-121	Barrier - Materials; Greaseproof, Flexible (Waterproof)
MIL-B-130	Barrier - Material; Paper Non-corrosive
MIL-A-148	Aluminum Foil
MIL-A-8625(ASG)	Anodic Coatings; (for) Aluminum and Aluminum alloys
MIL-P-18058	Preparation for Delivery of Machine Tools for Domestic and Overseas Shipment

STANDARDS

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

PUBLICATIONS

Air Force - Navy Aeronautical Bulletin

No. 143 Specifications and Standards; Use of

2.2 Availability of Documents.-

2.2.1 When requesting specifications, standards, or drawings, refer to both title and number and latest revision and/or amendment.

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2.2.2 Copies of this specification and applicable specifications, standards and drawings may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Attention: Code ODPT, Philadelphia 11, Pennsylvania.

2.3 Other Publications.-

2.3.1 The following document forms a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

NATIONAL MACHINE TOOL BUILDERS' ASSOCIATION
Machine Tool Electrical Standards

2.3.1.1 Copies of NMTBA Electrical Standards may be obtained from the National Machine Tool Builders' Association, 10525 Carnegia Avenue, Cleveland 6, Ohio.

3. REQUIREMENTS

3.1 General Requirements.- The surfacer shall be suitable for dry sanding or grinding metals, plastics, glass and similar materials by means of an endless abrasive belt or disc.

3.2 Materials.-

3.2.1 Selection of Materials.- Specifications and standards for all materials, parts and Government certification and approval of processes and equipment which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph:

3.2.2 Standard Parts.- AN or MIL standard parts shall be used wherever they are suitable for the purpose and shall be identified by their part numbers. Commercial utility parts, such as screws, bolts, nuts, cotter pins, et cetera, may be used provided they have suitable properties and are replaceable by the AN or MIL standard part without alteration and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts lists. In application for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

3.2.2.1 The materials used in the construction of the surfacer and accessory equipment shall be of high quality and suitable for the intended purpose to insure a long trouble free life of the equipment.

3.3 Design.-

3.3.1 General.- The surfacer shall be of the latest commercially available design of the manufacturer. Figure 1 herein is descriptive and non-restrictive and is not intended to preclude the purchase of surfacers otherwise conforming to this specification.

3.3.1.1 The surfacer shall be of rugged and substantial construction. It shall be free of vibration, provide for easy and convenient operation and shall afford safety to the operator.

3.3.2 Interchangeability.- All similar items furnished under a contract or order and fabricated under the same plans, shall be interchangeable with and shall not require any machining or hand fitting to the equipment at time of assembly.

3.3.3 Frame.- The frame shall be of semi-steel casting, rigidly reinforced to eliminate vibration, accurately machined for alignment of drive shafts, pulleys, work tables and the sanding disc. The base shall have provisions for mounting of safety guards for the drive pulleys and the sanding disc.

3.3.4 Abrasive Belt Pulleys.- The abrasive belt pulleys shall be approximately five (5) inches in diameter and of a width suitable for the utilization of a four (4) inch wide abrasive belt. The pulleys shall be accurately machined and balanced.

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The abrasive belt drive pulley shall be rubber faced and slightly crowned to assure accurate tracking of the abrasive belt. The pulleys center to center distance shall be a minimum of 18 inches.

3.3.4.1 The idler abrasive belt pulley shall have provisions for adjusting the tracking and tension of the abrasive belt.

3.3.4.2 Power for the abrasive belt drive shall be transmitted from the motor to the drive pulley by "V" type belts.

3.3.5 Abrasive Belt Tables.- The abrasive belt shall be equipped with a support table and a stop table. The belt support table shall be approximately twelve (12) inches by four and one quarter (4-1/4) inches and shall be located as to provide support to the upper belt surface; shall be of a cast construction with the surface in contact with the abrasive belt machined smooth and shall be adjustable in height. The stop table shall be approximately eight and one half (8-1/2) inches by six (6) inches; shall be of cast construction, machined smooth and shall have a mitre gauge slot. The stop table shall be mounted directly above the upper surface of the abrasive belt sanding surface, and shall be located to provide the maximum utilization of the sanding table surface. The stop table shall be of the adjustable angle type with a minimum adjustment of 45 degrees from the normal position of perpendicular to the abrasive belt. The adjustable angle shall be indicated on a graduated quadrant.

3.3.6 Disc.- A sanding disc shall be provided approximately ten (10) inches in diameter; shall be of cast construction, accurately balanced and faced to receive abrasive discs, mounted on the shaft of the abrasive belt drive pulley.

3.3.7 Disc Sanding Table.- A tilting hinged table approximately twelve (12) inches by six (6) inches shall be provided; shall be cast construction and shall have a smooth precise finish with a mitre gauge slot. The table shall be hinged to provide easy access to the abrasive disc and tilt from a position perpendicular to the disc to a minimum of 45 degrees from this position. The tilt angle shall be indicated on a graduated quadrant.

3.3.8 Stand.- The surfacer shall be mounted on a rigid steel stand approximately thirty-one (31) inches high, twenty-one and three-quarters (21-3/4) inches wide and thirty-two (32) inches long having an adjustable work shelf, adjustable motor brackets and steel shelf.

3.3.9 Bearings.- Ball bearings shall be of the anti-friction type, dust sealed, and grease packed so as to require no further lubrication during the life of the machine.

3.3.10 Speed.- Three speeds, approximately 760, 1350 and 2275 RPM shall be provided for the drive shaft.

3.4 Electrical Equipment.- All electrical equipment shall conform to NMTBA Electrical Standards and shall be furnished for operation on an alternating current 220 Volt, 60 Cycle, 3 Phase power source.

3.4.1 Motor.- The motor shall be one-half (1/2) horse power, enclosed, ball bearing, 40 degree centigrade temperature rating, for operation on the current specified in paragraph 3.4.

3.4.2 Controls.- A push button switch conveniently located to the operator shall be provided for the motor.

3.5 Equipment.- The surfacer shall be furnished complete with all the necessary equipment ready for operation. The following abrasive belts and discs shall be furnished with each machine:

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3 belts	36 grit
3 belts	60 grit
3 belts	120 grit
2 packages, 6 each,	40 grit discs
2 packages, 6 each,	60 grit discs
2 packages, 6 each,	80 grit discs

3.6 Finishes and Protective Coatings.- All protected metal parts and surfaces that are constructed from corrosion resistant material and on which a protective coating would be detrimental to proper functioning, need not be given protective coatings. All exposed metal parts and surfaces, and all protective metal parts on which a protective coating will not be detrimental to proper functioning shall be treated to satisfy the requirements of the specification.

3.6.1 Aluminum Alloy Parts.- Aluminum alloy surfaces shall be anodized in accordance with Specification MIL-A-8625(ASG) and then given two (2) coats of enamel conforming to Specification TT-E-485C.

3.6.2 Ferrous Parts.- Ferrous part surfaces shall be properly cleaned and painted with two (2) coats of enamel conforming to Specification TT-E-485.

3.6.3 Color.- The color of the surfacer shall be equivalent to color 2410 or color No. 2625, of Specification TT-C-595.

3.7 Identification of Product.- The surfacer shall be marked for identification in accordance with Standard MIL-STD-130, also the unit will be marked with the date of acceptance and length of the warranty.

3.8 Workmanship.- The workmanship shall be equal in every respect to the best modern commercial practice. This shall include accuracy, finish, fit of mating parts, and general appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests.- The inspection and testing of the surfacer shall be classified as acceptance tests.

4.2 Acceptance Tests.-

4.2.1 Individual Tests.- Each surfacer shall be subjected to the following tests:

4.2.1.1 Examination of Product.- The surfacer shall be inspected to determine compliance with the requirements specified herein with respect to material, workmanship and marking.

4.2.1.2 The surfacer shall be subjected to such tests as may be desired by the procuring activity to determine compliance with this specification.

4.2.1.3 Rejection.- Rejected articles shall not be resubmitted for approval without the inspector giving full particulars concerning previous rejection.

4.2.2 Test Reports.- Test reports of electrical equipment are not required. The contractor for the complete machine shall furnish a certificate by the electrical manufacturer, stating that the electrical equipment conforms with the requirements of this specification. If requested by the Naval Inspector, certified copies of the manufacturer's test reports shall be made available at the place of manufacture of the complete machine for his information and guidance.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.-

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5.1.1 Immediate Use (for domestic shipment with immediate use at initial destination). - The surfacer shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.1.2 Limited Storage (for domestic shipment with maximum storage of 180 days). The surfacer shall be preserved and packaged in accordance with the manufacturer's commercial practice. Packaging material in direct contact with unprotected surfaces susceptible to damage by corrosion shall conform to Specifications MIL-B-121, MIL-B-130 or MIL-A-148.

5.1.3 Extended Storage (for overseas shipment and domestic shipment with more than 180 days storage).- The surfacer shall be preserved and packaged in accordance with Military Specification MIL-P-18058.

5.2 Packing.-

5.2.1 Immediate Use (for domestic shipment with immediate use at initial destination).- Packages which require overpacking for acceptance by the carrier shall be packed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall conform to Consolidated Freight Classification Rules or regulations of other common carriers as applicable to the mode of transportation.

5.2.2 Domestic Shipment.- Unless otherwise specified, the preserved unit shall be prepared for domestic shipment in accordance with good commercial practice and acceptable to the carrier.

5.2.3 Overseas Shipment.- Surfacers preserved and packaged as specified in paragraph 5.1.3 shall be packed in accordance with MIL-P-18058.

5.3 Marking.- Interior and exterior containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering Data.-

6.1.1 Specification Identification.- Requests, requisitions, schedules and contracts or orders must contain the title of the specification, the number and the date.

6.1.2 Content.- Requests, requisitions, schedules and contracts or orders should cover the following features:

(a) Specifications or descriptions of desired extra attachments and equipment other than that ordinarily supplied with the machine.

6.2 Bidders shall submit with their bids, cuts or illustrations and complete description, all in duplicate of the article it is proposed to furnish, also cuts or photographs of attachments that may be specified, together with a description of each.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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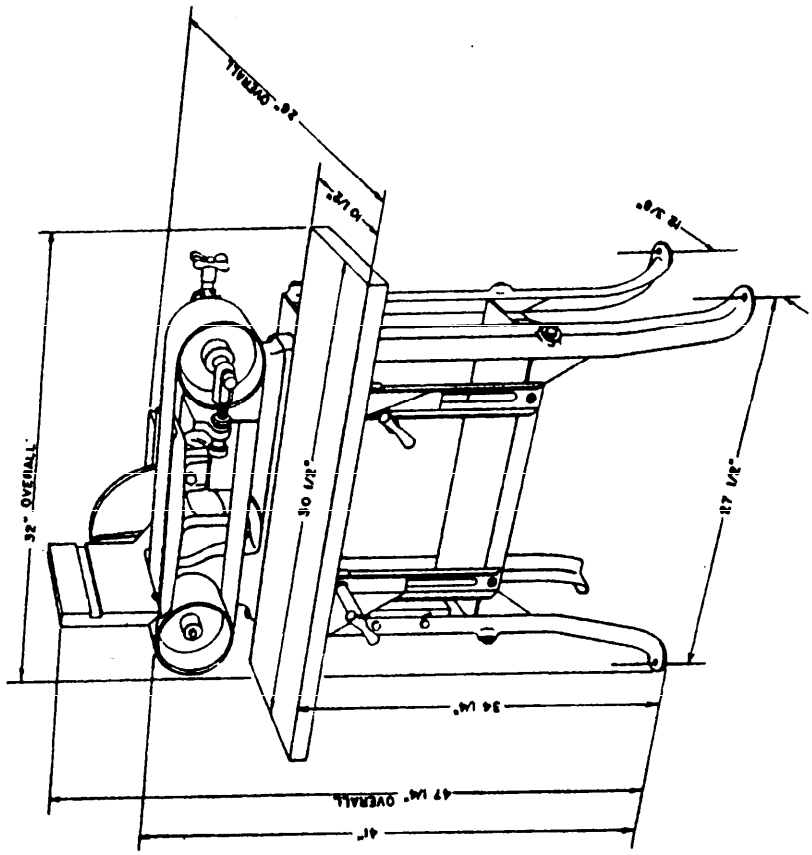


FIGURE 1 - SANDER

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